

AI Usage Declaration

Project: Super Mario Game (CS202 Final Project) **Group:** 52 **Members:** - Member A: Huy Nguyen (Engine & Infrastructure) - Member B: Partner (Entities & Gameplay)

Statement of Usage

We declare that Artificial Intelligence (AI) tools were utilized during the development of this project. The AI acted as a pair-programming assistant, contributing to the following areas:

1. **Architecture & Design:** Suggesting and refining structural design patterns (e.g., State, Observer, Factory, Singleton, Memento, Object Pool).
2. **Implementation:** Assisting in writing boilerplate C++ code, engine subsystems, algorithms (e.g., AABB collision), and gameplay logic.
3. **Debugging & Refactoring:** Identifying bugs, fixing logic flaws, and optimizing code structure for maintainability.
4. **Documentation:** Generating report drafts, logging task history, and formatting class diagrams.

Tools Used

- Gemini / Claude via Antigravity / AgentHub for agentic AI coding.

Academic Integrity

While AI assisted in the development, we carefully reviewed, tested, and integrated all AI-generated code. We understand the underlying logic of the submitted system and take full responsibility for its correctness, architecture, and behavior. The custom game logic, level design, and system integration were actively directed, guided, and verified by the human developers.

Signatures:

_____ (Huy Nguyen)

_____ (Partner)

Date: August 30, 2026